

Chikungunya transmission and population susceptibility in Ceará, Brazil

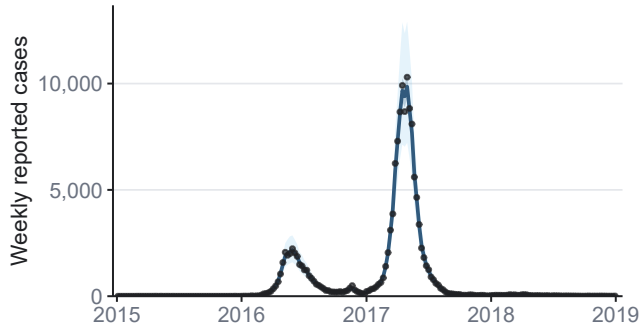
Bayesian renewal model, weekly observations from 2015 to 2018

A

Reported cases and posterior prediction

Shading: 95% predictive interval; dark band: 50% CrI

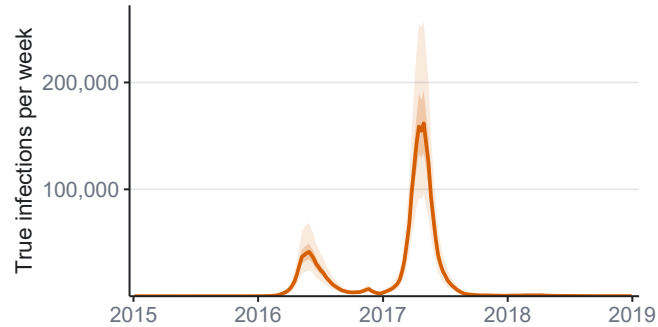
— Model expectation • Observed cases



B

Latent infection incidence

Median, 50% and 95% credible intervals

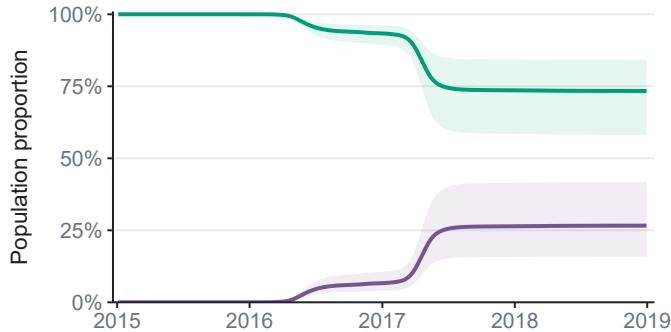


C

Population susceptibility and immunity

Closed population; lifelong immunity

— Infection-derived immune — Susceptible

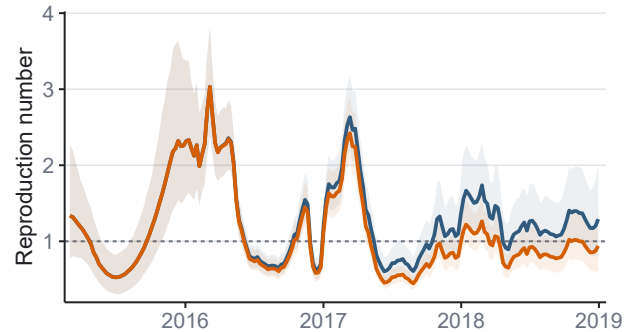


D

Time-varying reproduction numbers

$$R_{\text{eff}}(t) = R_0(t) \times S(t-1)/N$$

— $R_0(t)$ — $R_{\text{eff}}(t)$

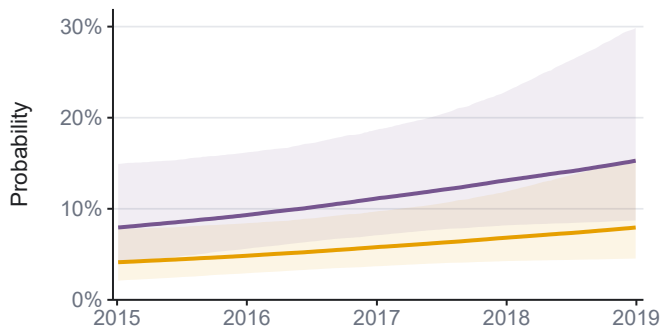


E

Case ascertainment

External symptomatic proportion fixed at 0.52

— Infection-to-report — Symptomatic-case reporting



F

Cumulative infection and serological anchor

Yellow field: June-December 2018 survey

— Ceara — Juazeiro model — Juazeiro observed

